

KANWAL BOPARAI

kgsbopar@uwaterloo.ca | +1 437-477-1085 | github.com/KanwalBoparai | linkedin.com/in/kgsbopar | Portfolio

Full-stack engineer and founder shipping production AI products. Computer Science at Waterloo (CS '27). Comfortable across TypeScript, Python, LLMs, and the modern web stack; bias toward ownership and zero-to-one builds.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C/C++, SQL, HTML/CSS, Liquid

Frontend: React, Next.js, Tailwind CSS, shadcn/ui, Three.js / React Three Fiber, Framer Motion, React Hook Form, Zod, Ag-Grid, Figma

Backend & Data: Node.js, FastAPI, GraphQL (Apollo), RESTful APIs, JWT Auth, PostgreSQL, Supabase, SQLite, SQLAlchemy, Vercel Blob, Google Calendar/Sheets APIs, AWS, GCP, Vercel

AI / LLMs: OpenAI & Anthropic APIs, Claude Code, OpenRouter, Vapi (Voice Agents), Speech-to-Text/Text-to-Speech, Hugging Face, LangChain, Prompt Engineering, Structured Outputs, Function Calling & Tool Use, RAG, AI Agents, Evals & Guardrails

DevOps & Tooling: Git, GitHub Actions (CI/CD), Codespaces, Jest, React Testing Library, Postman, pnpm, Agile/Scrum

EXPERIENCE

Wurth Canada Limited, IT Analyst

Jan 2026 – Apr 2026

Guelph, ON

- Supporting MDM and Azure AD migration: redesigning device enrollment, compliance policies, and access workflows with RAG-style retrieval for internal IT documentation.
- Partnering with Würth Germany on an AI governance platform that restricts unauthorized AI usage and prevents company data leakage to non-approved models.
- Leading upgrade and stabilization of the Speedy mobile app for new iOS releases across MDM policies, API integrations, and backend services.

Modern Automotive Limited, Software Engineering Intern

Oct 2024 – Dec 2024

Remote

- Built full-stack features for a hybrid-vehicle control dashboard in React (TypeScript) and Node.js, owning RESTful APIs, JWT auth, and third-party integrations for real-time data sync end-to-end.
- Authored automated tests with Jest and React Testing Library; designed a CI/CD pipeline with GitHub Actions that reduced deployment time by 40%.
- Shipped features in two-week Agile sprints with daily stand-ups, owning work from design through production deployment.

CAA (Canadian Automobile Association), IT Support

Sep 2023 – Dec 2023

Toronto, ON

- Built a Python + Hugging Face Speech2Text pipeline that converted call recordings into searchable transcripts; automated reporting pipelines, reducing manual effort by 30%.

BAAJ, Founder & Engineer

Dec 2022 – Jun 2023

Canada

- Founded and ran a drop-shipping e-commerce business solo, generating real revenue from day one before pausing to focus on university.
- Built the full storefront using Shopify Liquid, HTML/CSS, and JavaScript; designed prototypes and UI in Figma with focus on responsive layout, typography, and visual hierarchy.
- Drove growth using Google Analytics and Facebook Insights to optimize conversion; negotiated with suppliers and trained customer service team.

Crystal Hood System, Web Developer (Freelance)

Feb 2021 – Aug 2021

Toronto, ON

- Built and shipped a production marketing site with TypeScript and Next.js focused on responsive design and SEO; integrated Firebase, next-seo, and Google reCAPTCHA.

PROJECTS

Delegate-AI | React, TypeScript, FastAPI, LLMs

Live

- Voice-controlled Chrome extension that translates natural language into multi-step browser actions across any site using LLM-powered agent workflows.
- Architected the full-stack agent system with embeddings-based retrieval, structured prompting, and function calling; built RAG pipelines, guardrails, and an eval framework to reduce hallucinations and ensure safe, low-latency execution.

ApplyPilot | Next.js, TypeScript, PostgreSQL, LLMs

GitHub

- Full-stack AI job application platform with LLMs via OpenRouter (Llama 3.1) powering resume parsing, outreach email generation, and job tracking.
- Shipped a Chrome extension that scrapes job postings from any page and syncs in real time; structured-prompting pipelines for schema-conformant LLM outputs. Deployed serverlessly on Vercel.

Patient Intake Portal | Next.js, GraphQL, Supabase, TypeScript

Live

- Built a full-stack 3-step patient intake flow (demographics → documents → consent) with an operational admin dashboard; one React-Hook-Form + Zod schema validates every step, document uploads go to Vercel Blob over REST while patient data flows through a typed Apollo GraphQL API, and record atomicity is enforced via resolver-side Zod re-validation and Postgres NOT NULL/CHECK constraints, with records surfaced in Ag-Grid (sort, filter, in-app preview).

Carpet Voice Agent | Python, FastAPI, Claude (Anthropic), Vapi

GitHub

- Outbound AI voice agent that calls customers about a promotion and books in-home appointments via an async Claude (Anthropic) tool-use loop with three live tools — Google Calendar availability/booking and Google Sheets CRM logging — over Vapi telephony (STT/TTS), with strict system-prompt guardrails and graceful degradation keeping it on-script and factual.

3D Portfolio | React, Three.js, React Three Fiber, Tailwind

Live

- Fully interactive 3D portfolio with canvas geometry, Framer Motion animations, and a modular component architecture; designed every layout and transition by hand.

EDUCATION

University of Waterloo, Bachelor of Computer Science

Jan 2023 – 2027

- President's Scholarship of Distinction; top 15 percentile, JEE Main (NTA). Coursework: Algorithms, Data Structures, Machine Learning, Statistics, Linear Algebra, Database Systems.